Fiscal Year:	FY 2020	**Task Last Updated:**	FY 11/08/2020
PI Name:	Blakely, Eleanor A Ph.D.		
Project Title:	Simulation of GCR-Induced Harderian Gland and Lung Tumorigenesis		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
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Flight Assignment:	NOTE: End date is 2/4/2021 per PI (Ed., 11/7/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mao, Jian-Hua Ph.D. (Lawrence Berkeley National Laboratory) Chang, Polly Ph.D. (SRI International)		
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Task Description:	Increased cancer risk remains one of the primary concerns for travel into deep space and may preclude crewed missions to Mars due to the large uncertainties that currently exist in estimating cancer risk from the spectrum of radiation types found in space with available human epidemiological data on radiation-induced cancer. Existing data on human risk of cancer from exposures to X- and gamma-rays must be scaled to the many types and fluxes of radiations found in space using radiation quality factors and dose-rate modification factors. Linearity of dose-response is often assumed since the shape of the dose-response at doses below 100 mSv is unknown. However, preliminary analysis of our recently completed experiments with silicon and titanium ion suggest non-linear structure in the low-dose region for both Harderian and lung tumorigenesis. The goal of this proposal is to reduce uncertainties in the estimation of particle radiation carcinogenesis from galactic cosmic radiations (GCR). Our approach is to use the existing data on Harderian gland tumorigenesis from investigations of individual high-atomic number, and high-energy (HZE) ion beams to predict outcome of mixed ion chronic GCR exposures, and then measure Harderian gland and lung dose-dependent tumor data for such exposures. We will compare some of the measured cancer prevalence outcomes of a GCR-simulated exposure with the theoretical predictions based on experiments with individual ion beams. Our main hypothesis is that HZE carcinogenic effects are LET- (linear energy transfer) and dose-dependent and not additive at low particle doses due to the contribution of non-targeted effects. Completion of the proposed GCR studies using a mouse strain that has already been well characterized with a low spontaneous tumor background will allow a significant test of modeling approaches used to estimate carcinogenesis risk from a simulated GCR beamline.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Completion of our research on predicting and measuring HZE tumorigenic effects to test the hypothesis that HZE carcinogenic effects and LET- and dose-dependent are not additive at low particle doses due to the contribution of non-targeted effects will allow a significant test of modeling approaches used to estimate carcinogenic risk from single-ion and mixed radiation fields. This work will also benefit estimation of risk of tumorigenesis from mixed radiation fields on Earth.
Task Progress:	Recent experimental milestones: A. THREE HEAVY ION BEAM STUDY A total study cohort of ~ 3-month old 419 female CB6F1 mice (including 52 sham-treated controls) were shipped to Brookhaven National Laboratory (BNL). Animals were sham-treated or exposed to a 3- heavy ion mixed beam (260 MeV/u Si, 1 GeV/u Ti, and 600 MeV/u Fe) during NASA Space Radiation Laboratory (NSRL) 19A delivered in rapid sequence to a total dose of 30 cGy, or 60 cGy. After irradiation, animals were shipped to the vivarium at Lawrence Berkeley National Laboratory (LBNL) for 16-month long-term housing and maintenance, until their scheduled necropsy at SRI International on the first two weeks in August 2020. Tumor tissues were collected and fixed for potential down-stream analysis. Harderian Gland, lung tissues, and other tissues with gross abnormalities were collected, fixed, and processed to slides for histological evaluations. Other major organs such as kidneys, liver, lung, spleen, and brain were frozen and fixed for downstream processing or for sharing with other NASA investigators. Samples derived from bone marrow from 10 animals in each group were sent to Dr. Megumi Hada's laboratory for fluorescent in situ hybridization (FISH) staining with chromosomes 1, 2, and 4 whole chromosome probes to determine long-term aberration prevalence in these animals. Slides of colon tissues derived from 10 animals in each group were shipped to Dr. Jerry Shay's Laboratory for colon polyp assessment. Fixed ovarian tissues were to Dr. Ulrike Lederer's Laboratory for analysis. Sample analysis work is in progress and results will be reported in the next progress report. B. SIX-ION GCRsim STUDY We irradiated animals to a single total dose of 100 cGy of GCRsim in the Spring of 2018. The irradiation procedures were described in the 2019 progress report. The study design is summarized briefly here. Protons 250 MeV-60cGy (60%) + Helium 228 MeV/u-20 cGy (20%) + Oxygen 350 MeV-10 cGy (10%) + Silicon 260 MeV/u-2.5 cGy (2.5%) + Titanium 1 GeV/u-2.5 cGy (2.5%) + Iron 600 MeV/u-5 cGy (5%) This 6-ion combination is compliant with the requirements stated in the NASA Research Announcement where protons will contribute to 50%-60% of the total dose, 10%-20% of the dose contribution will be from alphas, 5%-10% of the contribution will be from HZE ions of 3>Z>9, and 5%-10% of the contributing dose will be from HZE ions of Z>10. B.1 Chromosome Aberrations: One of the key observations from the analysis of murine chromosome aberrations from the bone marrow of the mice harvested at 16 months post-irradiation is the significantly increased heterogeneity in chromosome exchanges/100 cells in the mice exposed to the 6-ion beams (p=0.007 by Fisher's exact test), compared to the sham-treated controls. A significant increase in simple exchanges (p=0.007 by Fisher's exact test), while no significant increase in complex exchanges (p=0.74 by Fisher's exact test) were found after exposure to the 6-ion beams, as well as the variability between individual mice. These observations will require follow up in order to confirm our speculation that the cells with complex exchanges are diminished in significance, due to their loss during the 16-month period post-treatment. B.2. Synergy Modeling of the 6-Ion Beam GCRsim Results: We completed theoretical modeling of the data obtained in the 6-ion beam GCRsim Harderian Gland tumor prevalence study, to test for synergy or antagonism. A modern synergy theory, Incremental Effect Additivity (IEA) was used, for the reasons detailed in Huang, EG, RY Wang, L Xie, P Chang, G Yao, B Zhang, DW Ham, Y Lin, EA Blakely and RK Sachs (2020). "Simulating galactic cosmic ray effects: Synergy modeling of murine tumor prevalence after exposure to two one-ion beams in rapid sequence." Life Sci Space Res (Amst) 25: 107-118 and its web supplement. The 6-ion beam exposure with a total dose of 100 cGy was measured to yield a Harderian Gland tumor prevalence of 23.5% above background.

	We used synergy theory analysis to estimate the IEA prediction and confidence intervals (CI) for the prevalence if no synergy or antagonism is present among the components of the mixed beam. Our analysis assumed that for HZE non-targeted effect, action dominates at very low doses. The 95% CI was calculated appropriately, taking adjustable parameter correlations into account. Incorrectly neglecting parameter correlations gives a substantially wider 95% Ci and leads to an unduly pessimistic estimate of the upper 95th percentile height that NASA emphasizes. We compared the IEA prediction with the data. We found that the measured prevalence was slightly higher than the predicted prevalence but there was no significant synergy or antagonism among the components of the six-ion mixture. This result was consistent with our expectations.
Bibliography Type:	Description: (Last Updated: 02/12/2026)
Abstracts for Journals and Proceedings	Chang P, Bakke J, Grover A, Rusek A, Snyder D, Rosen C, Bjornstad KA, Mao, JH, Sachs RK, Blakely EA. "High-LET Induced Lung Tumorigenesis." Presented at the 65th Annual Meeting of the Radiation Research Society, San Diego, CA, November 3-6, 2019. Online Proceedings. 65th Annual Meeting of the Radiation Research Society, San Diego, CA, November 3-6, 2019. , Nov-2019
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